Virtual University	Final-term EXAMINATION SEMESTER Fall 2004 STA-301 Statistics and probability(Paper-1)	Total Marks:50 Duration:120min

Student ID / Login ID.	
Name.	
PVC Name /Code	
Date	

Maximum Time Allowed: (2 Hours)

Please read the following instructions carefully before attempting any of the questions:

1. **Attempt all questions. Marks are written adjacent to each question.**
2. **Do not ask ay questions about the contents of this examination from anyone.**
   a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
   b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
   c. **Write all steps, missing steps may lead to deduction of marks.**
3. **You are allowed to use the calculator & Statistical tables in order to solve the questions.**
4. **For your convenience we are providing you the following symbols,**

$\square$, $\cap$, $\bar{X}$ ‘ $t_{\alpha/2(\nu)}$, $t_{(\nu)}$, $\mathbb{R}_1$, $\mathbb{R}_2$, α, α_1, α_2 , α_3 , α^4, α_2 , α_3^2 , $\int_2$, $\int_4$, $\square x$, $\square y$, $\square xy$, $\square x_2$, $\square^{+\square}_{\square\square} y_2$,

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Questions	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Marks								

Questions	Q9	Q10	Q11	Total
Marks				

Question No: 1 **Marks:1**

If mean=median=mode then curve is symmetric.

- o True
- o False

Question No: 2 **Marks:1**

Distribution Function is also called the step function.

- o True
- o False

Question No: 3 **Marks:1**

In the case of sampling with replacement as well as in the case of sampling without replacement, we have:

$\mu_{\bar{X}} = \mu$

- o True
- o False

Question No: 4 **Marks:1**

The rejection region is also called the critical region.

- o True
- o False

Question No: 5 **Marks:4**

Describe the continuous and discrete random variable with example.

Question No: 6 **Marks:4**

Define and describe the Least Significance Difference (LSD).

Question No: 7 **Marks:4**

Write down the properties of normal distribution.

Question No: 8 **Marks:4**

Discuss on null and alternative hypothesis.

Question No: 9 **Marks:10**

Following is given the data of Traffic Density and Accident rate.

Traffic Density	30	35	40	45	50	60	70	80	90
Accident rate	2	4	5	5	8	15	24	30	32

a. Plot the data on scatter dia-gram

b.Calculate the correlation co-efficient between traffic density and accident rate

c.Interpret the correlation.

Question No: 10 **Marks:10**

A continuous random variable X that can assume values between x=1 and x=4 has a density function given by

$f(x) = \frac{1}{3}$

a. Show that the area under the curve is equal to 1.

b.Find $p(1.5 < X < 3)$

c. $P(X \leq 2.2)$

Question No: 11 **Marks:10**

Let X and Y be two discrete r.vs with the following joint p.d.

x **y**	2	4
1	0.1	0.15
3	0.2	0.30
5	0.1	0.15

Find $E(X)$ and $E(Y)$.

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